Clustering

Clustering

- Unsupervised learning
- No labelled data
- Clustering is the process of grouping a set of data objects into multiple groups or clusters so that objects within a cluster have high similarity, but are very dissimilar to objects in other clusters.
- Dissimilarities and similarities are assessed based on the attribute values describing the objects and often involve distance measures.

Methods

- Partition based methods:
  - Given a set of n objects, a partitioning method constructs k partitions of the data, where each partition represents a cluster and k <= n. That is, it divides the data into k groups such that each group must contain at least one object.
  - Most partitioning methods are distance-based.
    - Kmeans
    - Kmedoids
- Hierarchical methods
  - Creates a hierarchical decomposition of the given set of data objects.
  - Hierarchical clustering methods can be distance-based or density- and continuity based.
  - Hierarchical methods suffer from the fact that once a step (merge or split) is done, it can never be undone.
    - Agglomerative
    - Divisive

- Density based Methods
  - The general idea is to continue growing a given cluster as long as the density (number of objects or data points) in the “neighborhood” exceeds some threshold.
  - For example, for each data point within a given cluster, the neighborhood of a given radius has to contain at least a minimum number of points.
  - Such a method can be used to filter out noise or outliers and discover clusters of arbitrary shape.
    - DBSCAN

• Grid Based Methods

- Grid-based methods quantize the object space into a finite number of cells that form a grid structure.
- All the clustering operations are performed on the grid structure (i.e., on the quantized space).
- The main advantage of this approach is its fast processing time, which is typically independent of the number of data objects and dependent only on the number of cells in each dimension in the quantized space.
- Using grids is often an efficient approach to many spatial data mining problems, including clustering. Therefore, grid-based methods can be integrated with other clustering methods such as density-based methods and hierarchical methods.

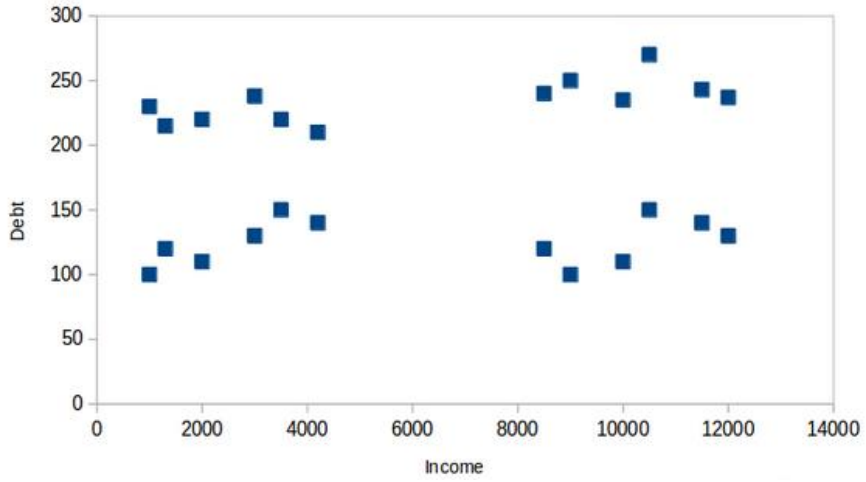
Method	General Characteristics
Partitioning methods	– Find mutually exclusive clusters of spherical shape – Distance-based – May use mean or medoid (etc.) to represent cluster center – Effective for small- to medium-size data sets
Hierarchical methods	– Clustering is a hierarchical decomposition (i.e., multiple levels) – Cannot correct erroneous merges or splits – May incorporate other techniques like microclustering or consider object "linkages"
Density-based methods	– Can find arbitrarily shaped clusters – Clusters are dense regions of objects in space that are separated by low-density regions – Cluster density: Each point must have a minimum number of points within its "neighborhood" – May filter out outliers
Grid-based methods	– Use a multiresolution grid data structure – Fast processing time (typically independent of the number of data objects, yet dependent on grid size)

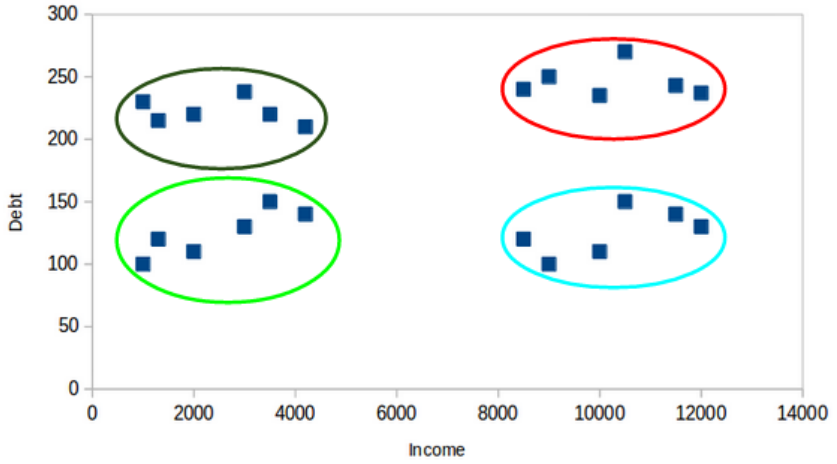
K-means Clustering Algorithm

K-means Clustering

- The objective of K-means: group similar data points together and discover underlying patterns. To achieve this objective, K-means looks for a fixed number (*k*) of clusters in a dataset.”
- A cluster refers to a collection of data points aggregated together because of certain similarities.
- You’ll define a target number *k*, which refers to the number of centroids you need in the dataset. A centroid is the imaginary or real location representing the center of the cluster.

- In other words, the K-means algorithm identifies *k* number of centroids, and then allocates every data point to the nearest cluster, while keeping the centroids as small as possible.
- The *'means'* in the K-means refers to averaging of the data; that is, finding the centroid.





Properties of Clusters

- **Property 1**
  - All the data points in a cluster should be similar to each other
  - Intra cluster similarity should be high

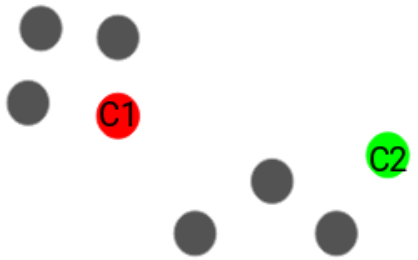


- **Property 2**
  - The data points from different clusters should be as different as possible
  - Inter cluster similarity should be low

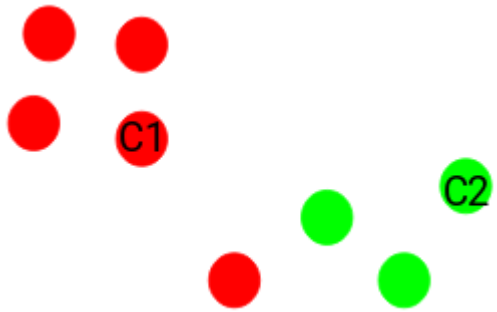
- ***The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid.***
- ***Steps:***
- Step 1: Choose the number of clusters *k:*The first step in k-means is to pick the number of clusters, k.



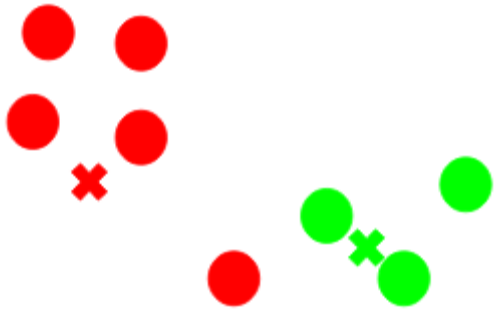
- Step 2: Select k random points from the data as centroids



- Step 3: Assign all the points to the closest cluster centroid



- Step 4: Recompute the centroids of newly formed clusters



- Step 5: Repeat steps 3 and 4



- **Stopping Criteria for K-Means Clustering**
- There are essentially three stopping criteria that can be adopted to stop the K-means algorithm:
  1. Centroids of newly formed clusters do not change
  2. Points remain in the same cluster
  3. Maximum number of iterations are reached

Example

- Use k means clustering to group the data points to 2 clusters. Use Euclidean measure.

x1	1	2	2	3	4	5
x2	1	1	3	2	3	5

- Step 1: chose 2 random points as initial cluster: C1= (2,1), C2=(2,3)
- Step2: Find the distance between every data points to the cluster centers

Data point	Distance from C1 (2,1)	Distance from C2 (2,3)	Assigned cluster
A1(1,1)	1	2.24	C1
A2(2,1)	0	2	C1
A3(2,3)	2	0	C2
A4(3,2)	1.41	1.41	C1
A5(4,3)	2.83	2	C2
A6(5,5)	5	3.61	C2

C1 {A1, A2, A4}

C2{A3, A5, A6}

Updated Centroid C1= (1+2+3)/3, (1+1+2)/3 = (2,1.33)

Updated Centroid C2= (2+4+5)/3, (3+3+5)/3= (3.67,3.67)

Step 3: Repeat step 2 until there is no change in centroids

C1 {A1, A2, A3, A4}

C2{A5, A6}

Q2

- Cluster the following eight points (with (x, y) representing locations) into three clusters:
  - A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)
- Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

Iteration 1:

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (5, 8) of Cluster-02	Distance from center (1, 2) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	5	9	C1
A2(2, 5)	5	6	4	C3
A3(8, 4)	12	7	9	C2
A4(5, 8)	5	0	10	C2
A5(7, 5)	10	5	9	C2
A6(6, 4)	10	5	7	C2
A7(1, 2)	9	10	0	C3
A8(4, 9)	3	2	10	C2

a. New clusters are-

Cluster-01:

A1(2, 10)

Cluster-02: Second cluster contains points-

- A3(8, 4)
- A4(5, 8)
- A5(7, 5)
- A6(6, 4)
- A8(4, 9)

Cluster-03: Third cluster contains points-

- A2(2, 5)
- A7(1, 2)

b. Re-compute the new cluster centers.

Cluster-01:A1(2, 10)
Cluster-02: ((8 + 5 + 7 + 6 + 4)/5, (4 + 8 + 5 + 4 + 9)/5)=(6,6)
Cluster-03: ((2 + 1)/2, (5 + 2)/2)= (1.5, 3.5)

Iteration-02: We calculate the distance of each point from each of the center of the three clusters and proceed as in iteration 1.

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (6, 6) of Cluster-02	Distance from center (1.5, 3.5) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	8	7	C1
A2(2, 5)	5	5	2	C3
A3(8, 4)	12	4	7	C2
A4(5, 8)	5	3	8	C2
A5(7, 5)	10	2	7	C2
A6(6, 4)	10	2	5	C2
A7(1, 2)	9	9	2	C3
A8(4, 9)	3	5	8	C1

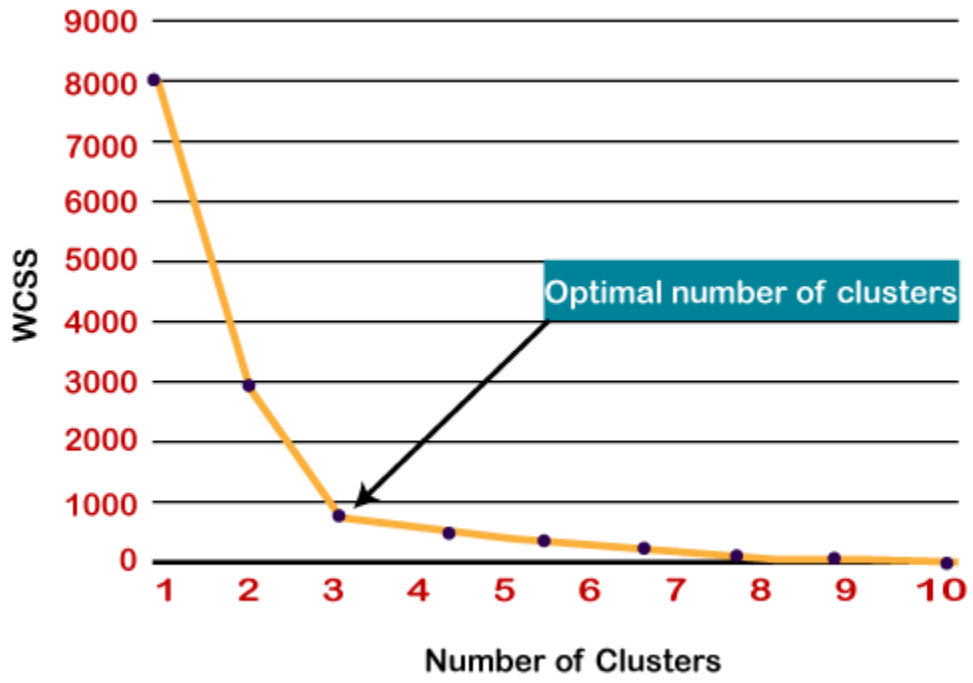
How to choose the value of "K number of clusters" in K-means Clustering?

- The performance of the K-means clustering algorithm depends upon highly efficient clusters that it forms. But choosing the optimal number of clusters is a big task.
- **Elbow Method**
  - The Elbow method is one of the most popular ways to find the optimal number of clusters. This method uses the concept of WCSS value. **WCSS** stands for **Within Cluster Sum of Squares**, which defines the total variations within a cluster. The formula to calculate the value of WCSS (for 3 clusters) is given below:

- WCSS= $\sum_{\text{Pi in Cluster1}} \text{distance}(P_i\ C_1)^2 + \sum_{\text{Pi in Cluster2}} \text{distance}(P_i\ C_2)^2 + \sum_{\text{Pi in CLuster3}} \text{distance}(P_i\ C_3)^2$

- $\sum_{\text{Pi in Cluster1}} \text{distance}(P_i\ C_1)^2$: It is the sum of the square of the distances between each data point and its centroid within a cluster1 and the same for the other two terms.
- To measure the distance between data points and centroid, we can use any method such as Euclidean distance or Manhattan distance.
- **To find the optimal value of clusters, the elbow method follows the below steps:**
- It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).

- For each value of K, calculates the WCSS value.
- Plots a curve between calculated WCSS values and the number of clusters K.
- The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.



Advantages and Disadvantages of K-means

Advantages

- It is very easy to understand and implement.
- If we have large number of variables then, K-means would be faster than Hierarchical clustering.
- On re-computation of centroids, an instance can change the cluster.
- Tighter clusters are formed with K-means as compared to Hierarchical clustering.

Disadvantages

- It is a bit difficult to determine the number of clusters i.e. the value of k.
- Output is strongly impacted by initial inputs like number of clusters (value of k).
- Order of data will have strong impact on the final output.
- It is very sensitive to rescaling. If we will rescale our data by means of normalization or standardization, then the output will completely change.
- It is not good in doing clustering job if the clusters have a complicated geometric shape.